

Theory of Knowledge Essay

Prescribed Title

To what extent do you agree with the claim that ‘any coincidence is worth noticing; you can throw it away later if it is only a coincidence’ (Agatha Christie). Answer with reference to mathematics and one other area of knowledge.

Student name: [Student name]

Candidate number: [Candidate number]

School: [School name]

Examination session: May 2027

Word count: 1,593 words

Contents

1. Introduction.....	1
2. Mathematics.....	1
2.1 Claim: coincidences generate mathematical questions.....	1
2.2 Counterclaim: patterns can deceive.....	2
3. Natural Sciences.....	3
3.1 Claim: anomalies can lead to discovery.....	3
3.2 Counterclaim: coincidences are cheap to find, costly to pursue.....	4
4. Conclusion.....	5
5. References.....	6

1. Introduction

In Agatha Christie's *Nemesis*, Miss Marple tells herself that any coincidence “is always worth noticing. You can throw it away later if it is only a coincidence” (Christie, 1971). For a detective this is sensible: noticing costs little, while missing a clue could let a murderer escape. As a claim about knowledge, it is more complicated. I take a “coincidence” to mean an apparent pattern or unexpected match that has, at first, no explanation. Calling it “worth noticing” means that treating it as a question, rather than ignoring it, helps to produce knowledge. But noticing is not knowing: a noticed coincidence has no justification yet. The quotation also assumes that the knower can later tell reliably when a pattern is “only a coincidence”, which depends on the methods available.

I will explore the claim in mathematics, where patterns are justified by proof, and in the natural sciences, where they rely on empirical evidence that is always probabilistic. I will argue that the claim is largely right about the value of noticing but too optimistic about throwing coincidences away. A coincidence is worth noticing when the knower has the expertise to recognise it as surprising and a reliable method to test it, but where data is produced on a huge scale, the word “any” becomes hard to defend.

2. Mathematics

2.1 Claim: coincidences generate mathematical questions

In mathematics the claim seems well supported, because many conjectures begin as patterns noticed before anyone can explain them. A striking example is “monstrous moonshine”. In 1978, John McKay noticed that 196,884, a coefficient of the j -function from number theory, is exactly one more than 196,883, the smallest dimension in which the Monster, a vast symmetry group from a separate field, can be represented non-trivially (Conway & Norton, 1979). The name “moonshine” reflected how far-fetched a link between these areas seemed. Instead of dismissing it, mathematicians found that later coefficients could also be built from the Monster's dimensions, and Conway and Norton (1979) turned these observations into precise conjectures, which Borchers (1992) eventually proved.

This example shows three things about mathematical knowledge. First, the coincidence was only noticeable to someone who knew both fields; to most people 196,884 is just a number, so what counts as “surprising” depends on the knower’s prior knowledge. Second, the coincidence itself justified nothing: until 1992 moonshine was treated as a conjecture, not as knowledge, because mathematics demands deductive proof. Third, even before a proof, there were reasonable criteria for taking the pattern seriously: it was exact rather than approximate, and it kept holding as more cases were checked. The coincidence directed attention; proof supplied justification.

2.2 Counterclaim: patterns can deceive

However, a noticed pattern can mislead precisely because it is persuasive. The more cases fit, the more it feels like knowledge, even though no number of cases proves a general statement. Guy (1988) called this the “strong law of small numbers”: “there aren’t enough small numbers to meet the many demands made of them”, so small cases often share patterns by accident.

The Borwein integrals show this clearly. The integral from 0 to infinity of $\sin(x)/x$ equals $\pi/2$. Multiplying in $\sin(x/3)/(x/3)$ still gives $\pi/2$, and so does adding $\sin(x/5)/(x/5)$, and so on up to $\sin(x/13)/(x/13)$: seven cases, all exactly $\pi/2$. But when $\sin(x/15)/(x/15)$ is included, the result falls short of $\pi/2$ by about 2.3×10^{-11} (Borwein & Borwein, 2001). Checked numerically to ten decimal places, this eighth case would also seem to “confirm” the pattern.

This challenges the claim in two ways. First, noticing can produce false confidence: a pattern that has worked seven times encourages confirmation bias, because each new case feels like evidence although it adds no proof. Second, Christie assumes we will know “later” whether something was only a coincidence, but in mathematics more observation cannot tell us this. The pattern was understood only through proof: the Borweins showed that the integrals equal $\pi/2$ exactly while the sum $1/3 + 1/5 + 1/7$ and so on stays below 1, and adding $1/15$ pushes it above 1. Even the failure, once explained, became knowledge.

Weighing these, the claim holds in mathematics, but only because proof acts as a powerful filter that can eventually settle, with certainty, whether a pattern holds. The danger lies not in noticing a coincidence but in mistaking accumulated examples for justification. A mathematical coincidence is worth noticing as a source of conjectures, not as a justification for them.

3. Natural Sciences

The natural sciences offer a useful contrast: they also search for patterns, but they depend on observation and cannot prove them.

3.1 Claim: *anomalies can lead to discovery*

Here, noticing seems even more important, because an unexpected observation is often the first sign that a theory is incomplete. In 1967, Jocelyn Bell, a Cambridge PhD student, noticed “a bit of scruff” on recordings from a new radio telescope (Bell Burnell, 1977). It proved to be pulses repeating every 1.337 seconds, so regular that the team nicknamed the source “LGM-1”, for “little green men”. Rather than dismissing it as interference, they tested it: the signal appeared four minutes earlier each day, keeping sidereal time (the timing of the stars), which ruled out a source on Earth (Hewish et al., 1968). When Bell found a similar signal elsewhere in the sky, an alien origin became implausible (Bell Burnell, 1977), and Gold (1968) proposed that the pulses came from rotating neutron stars.

This supports the claim, but the path to knowledge differs from mathematics. With moonshine, the numbers were certain and only their meaning was unclear; here, the first question was whether the pattern was even real. The coincidence became evidence by surviving attempts to explain it away, as each test eliminated a rival explanation. Justification came from ruling out alternatives and building an explanatory theory rather than from proof, so the resulting knowledge is strongly supported but revisable. Like McKay, Bell noticed the anomaly because she knew what ordinary interference looked like, so “worth noticing” again depends on the knower’s expectations. Here, the real risk would have been throwing the coincidence away too early.

3.2 Counterclaim: coincidences are cheap to find, costly to pursue

However, in modern science the word “any” becomes a problem, because when enough data is examined, coincidences are almost guaranteed. In December 2015, the ATLAS and CMS experiments at CERN’s Large Hadron Collider both reported more photon pairs than expected at about 750 GeV, a possible sign of a new particle. ATLAS found a “local” significance of 3.9 standard deviations, but after correcting for having searched across many energies, known as the “look-elsewhere effect” (Gross & Vitells, 2010), the “global” significance fell to about 2.1 (ATLAS Collaboration, 2016). Even so, around 500 theoretical papers tried to explain the bump. By August 2016, with much more data, it had disappeared and was accepted as a statistical fluctuation (“And So to Bed,” 2016).

In one sense, this is Christie’s advice working: the coincidence was noticed, investigated and thrown away. Yet it challenges the claim in three ways. First, the look-elsewhere effect is essentially the strong law of small numbers in scientific form: if you look in enough places, some bump will appear by chance, so not every one can be investigated. This is why particle physicists require five standard deviations before claiming a discovery (Lyons, 2013), a rule that decides in advance which coincidences deserve belief. Second, throwing it away was not free: hundreds of researchers spent months on noise, suggesting that the excitement of possible discovery can produce collective confirmation bias. Third, unlike a mathematical counterexample, the rejection was not final: the bump was not disproved but became very unlikely, judged against a threshold that is a community convention rather than a logical necessity. In science, “only a coincidence” is always a judgement of probability.

Weighing both examples, the claim holds in the natural sciences when a coincidence can be tested independently, through sidereal timing, a second source or more data. But “any” does not hold: science produces too many chance patterns to pursue them all. Unlike mathematicians, scientists cannot wait for proof; they must use probability to decide which coincidences deserve attention, and throwing one away is never completely final.

4. Conclusion

I agree with the claim to a considerable extent, but only if noticing a coincidence is clearly separated from accepting it as knowledge. In both areas, the coincidences behind moonshine and pulsars were worth noticing because they raised questions nobody had asked. Yet in both cases the coincidence justified nothing by itself: it became knowledge through proof in mathematics, and through eliminating rival explanations and building a physical theory in the natural sciences.

The counterclaims show that the second half of the quotation is weaker. “You can throw it away later” assumes a reliable method for doing so. Mathematics has one, since proof settles with certainty whether a pattern holds, although, as the Borwein integrals show, collecting examples cannot. The natural sciences can only discard coincidences provisionally, and with modern volumes of data they must filter coincidences before pursuing them. The value of a coincidence therefore lies less in the coincidence itself than in the expertise of the knower who recognises it and the methods available to test it.

This position has limits: my examples are well-documented cases, while many coincidences are quietly dropped without record, so it is hard to judge how often noticing pays off. Still, the analysis suggests a more accurate version of Christie’s claim: a coincidence is worth noticing when we have the means to test it and the willingness to throw it away.

5. References

- And so to bed for the 750 GeV bump. (2016, August 5). *Physics World*.
<https://physicsworld.com/a/and-so-to-bed-for-the-750-gev-bump/>
- ATLAS Collaboration. (2016). Search for resonances in diphoton events at $\sqrt{s} = 13$ TeV with the ATLAS detector. *Journal of High Energy Physics*, 2016(9), Article 1.
[https://doi.org/10.1007/JHEP09\(2016\)001](https://doi.org/10.1007/JHEP09(2016)001)
- Bell Burnell, S. J. (1977). Petit four. *Annals of the New York Academy of Sciences*, 302(1), 685–689. <https://doi.org/10.1111/j.1749-6632.1977.tb37085.x>
- Borcherds, R. E. (1992). Monstrous moonshine and monstrous Lie superalgebras. *Inventiones Mathematicae*, 109(2), 405–444. <https://doi.org/10.1007/BF01232032>
- Borwein, D., & Borwein, J. M. (2001). Some remarkable properties of sinc and related integrals. *The Ramanujan Journal*, 5(1), 73–89.
<https://doi.org/10.1023/A:1011497229317>
- Christie, A. (1971). *Nemesis*. Collins Crime Club.
- Conway, J. H., & Norton, S. P. (1979). Monstrous moonshine. *Bulletin of the London Mathematical Society*, 11(3), 308–339. <https://doi.org/10.1112/blms/11.3.308>
- Gold, T. (1968). Rotating neutron stars as the origin of the pulsating radio sources. *Nature*, 218(5143), 731–732. <https://doi.org/10.1038/218731a0>
- Gross, E., & Vitells, O. (2010). Trial factors for the look elsewhere effect in high energy physics. *The European Physical Journal C*, 70, 525–530.
<https://doi.org/10.1140/epjc/s10052-010-1470-8>
- Guy, R. K. (1988). The strong law of small numbers. *The American Mathematical Monthly*, 95(8), 697–712. <https://doi.org/10.2307/2322249>
- Hewish, A., Bell, S. J., Pilkington, J. D. H., Scott, P. F., & Collins, R. A. (1968). Observation of a rapidly pulsating radio source. *Nature*, 217(5130), 709–713.
<https://doi.org/10.1038/217709a0>
- Lyons, L. (2013). *Discovering the significance of 5 sigma* (arXiv:1310.1284). arXiv.
<https://doi.org/10.48550/arXiv.1310.1284>